How the Web Works

Lesson 6 · DNS, TCP/UDP, TLS & HTTP/1.1–2–3

In this lesson → ① Type a URL ② DNS ③ DNS as a tool ④ TCP vs UDP ⑤ TLS ⑥ HTTP versions ⑦ Request & response ⑧ Why it matters ⑨ Interview Q&A

Running project: SnapLink, our photo-sharing app that grows from 1 server to 100M users.

1 Type a URL: the journey



each step costs round trips → reuse connections, cache DNS

2 DNS: name → IP



answer cached with a TTL

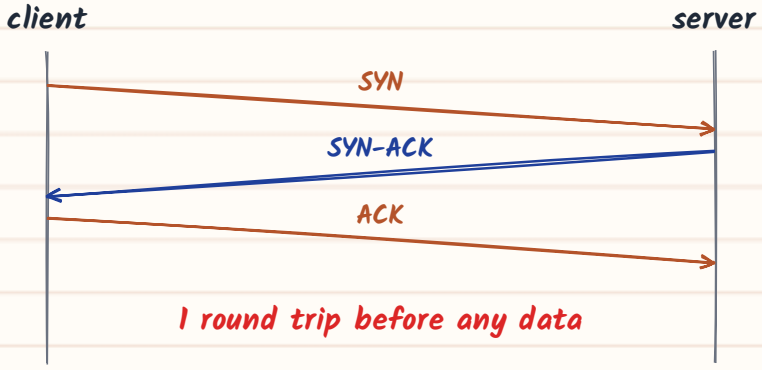
→ Resolver → root → TLD (.com) → authoritative

→ Cached at browser, OS, resolver for the **TTL**

→ Record types: A (IPv4), AAAA (IPv6), CNAME (alias), NS, MX

DNS technique	Use
Geo DNS / latency routing	send users to the nearest region
Multiple A records	simple load spreading
Short TTL	fast failover (but more lookups)
Health-checked DNS	remove dead endpoints

3 TCP vs UDP



	TCP	UDP
Connection	3-way handshake	none
Delivery	reliable, ordered, retransmits	best effort
Speed	slower	faster
Used by	HTTP/1.1–2, DBs, email	video calls, games, DNS, QUIC

4 TLS

- ✓ Agree on keys + verify server certificate
- ✓ TLS 1.3: 1 RTT handshake, 0-RTT resumption
- ✓ Terminate at load balancer / CDN edge to save work

5 HTTP versions



Version	Transport	Key idea
HTTP/1.1	TCP	one request at a time per connection; keep-alive
HTTP/2	TCP	multiplexed streams, header compression
HTTP/3	QUIC (UDP)	no TCP head-of-line blocking, faster setup

6 Request & response

Code	Meaning	Examples
2xx	success	200 OK, 201 Created, 204 No Content
3xx	redirect / cache	301, 302, 304 Not Modified
4xx	client error	400, 401, 403, 404, 409, 429
5xx	server error	500, 502 Bad Gateway, 503, 504

7 Why it matters in design

- ✓ Round trips = latency (US ↔ EU ≈ 150 ms each)
- ✓ Cache DNS; reuse connections (keep-alive, HTTP/2)
- ✓ Terminate TLS close to users
- ✓ Serve static content from a CDN (SD-08)

SYSTEM DESIGN NOTEBOOK

8 Interview Corner

Q. What happens when you type a URL?

A. DNS lookup → TCP handshake → TLS handshake → HTTP request → server response → browser renders (more requests for assets).

Q. TCP vs UDP?

A. TCP: reliable, ordered, connection-based. UDP: connectionless, fast, no delivery guarantees.

Q. What is DNS TTL?

A. How long a DNS answer may be cached; short TTLs enable faster failover.

Q. Why HTTP/2 and HTTP/3?

A. HTTP/2 multiplexes streams on one connection; HTTP/3 uses QUIC over UDP to avoid head-of-line blocking.

Q. Where is TLS terminated?

A. Usually at the load balancer or CDN edge, then traffic is re-encrypted or sent on a trusted network.

⚡ Revise in 60 seconds

- → **DNS**: name → IP, cached (TTL)
- → **TCP**: 3-way handshake
- → **UDP**: fast, no guarantees
- → **TLS 1.3**: 1 RTT
- → HTTP/2: streams
- → HTTP/3: QUIC over UDP
- → 2xx · 3xx · 4xx · 5xx
- → Fewer round trips = faster

Next lesson → SD-07
Load balancers: L4 vs L7, algorithms, health checks.
Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**